MScFE 652 PORTFOLIO MANAGEMENT

Group Work Project # 2

See grading rubric here

Scenario

Portfolio management is going through a renaissance. Why? Machine learning is changing the way people make decisions under uncertainty. Relying on time-tested, Nobel-Prize winning ideas sounds good, but in fact, these ideas may be outdated. Machine learning has already changed the way statistical analyses are done. When applied to the portfolio optimization process, that improvement could be enough to provide an advantage. Combined with leverage and other factors, these advantages could mean a substantial and significant outperformance over the competition.

Welcome to the 21st century of portfolio management! In this GWP, students will apply machine learning concepts of denoising, clustering, and backtesting to optimize a portfolio. Use the portfolios that were analyzed in the previous GWP.

Tasks

Groups of 3 members complete all the following steps:

Choose portfolios of at least 10 assets. Use January 1, 2024 - September 30, 2025 as the reference (training) period and use daily data.

Step 1. Determine the optimal Kelly portfolio. Can you find the optimal Kelly portfolio if no asset is allowed to be more than 20% of the total portfolio. If you can, find the optimal portfolio with such a constraint in place.

Step 2. Using data from October 1, 2025 to December 31, 2025, compare the Kelly portfolio with the double-Kelly and double-Kelly portfolio in terms of performance and risk (cumulative return, Sharpe ratio, Maximum Drawdown, 95%-cVaR). Organize your findings in a nice table so that the performance and risk of each are showing clearly. **NOTE**: *The model is to be estimated/trained using the data from January 1, 2024 to September 30, 2025 (in-sample period) and tested on the period October 1, 2025 to December 31, 2025 (out-of-sample period).*

Step 3. How would you backtest the model(s) in Step 1 and Step 2 using K-fold cross validation. Explain the process in half a page or so, then write the Python code to perform the backtesting and report the findings. (Use the entire 2 years of data from January 1 2024 to December 31, 2025 to do this Step).

Step 4. Then each student should explain (in no more than 2 pages per item) the features and benefits of each of these topics:

- Improvements using denoising
- Improvements using clustering. Ideally here you should discuss HRP (Hierarchical Risk Parity). If you use a different method, be sure to explain what you did. For example, groups could also opt for an easier to implement HEW (Hierarchical Equal Weight) portfolio or another method of your choice.
- Improvements using detoning.

Step 5. Students work together to apply the methodologies described in Step 4 to the portfolio used in Step 1 above. You can use third party software, but make sure that you explain the steps of what you are doing. In addition, students should prepare a document explaining the merits of applying one, two, or all of these “improvements” to the current portfolio. They could also theorize about what could be the best sequence with which to use the three tools listed above. (Be essential, do not “write a book”!)

Step 6. Students discuss if and how multiple improvements potentially outperform single improvements. Students show which combination of multiple improvements (if any) outperforms single improvements, according to at least three metrics of the group’s choice. Examples of metrics include return, Sharpe ratio, expected shortfall, variance, maximum drawdown, Sortino ratio, etc.)

Discuss if the ML methodologies helped the out of sample performance of the portfolio vs the more traditional approach using (some of the) metrics described in Step 4. You should do this by considering the following:

1. What are the differences in performance according to these metrics for the different combination of improvements?
2. What reasons might explain these differences in performances (with specific reference to the financial products, the historical data, the math of the model dynamics, etc.),
3. In which cases, if any, does the incremental gain in performance according to the chosen metrics justify the additional complexity and effort of the improvements?

Step 7: Preparing for a Technical Talk

1. Prepare 3 – 5 slides on a technical talk about how the portfolio methodology addressed and solved problems in 1 of the following 3 areas:
    a. Regularization
    b. Prediction vs. Interpretation
    c. Estimation Error
    d. Regime Shifts
2. Be sure to include definitions, equations, formulas about the term
3. Show a direct connection between the data you analyzed and the challenge selected.
4. Show a direct connection between the method you used and how it addressed the financial challenge (Note: Methods may not necessarily help solve the problems caused by the specific challenge)

Groups of 2 members:

1. Select 2 items only from Step 4;
2. Students complete Steps 1, 2, 3, 4, 5 and 6. They do NOT do Step 7.

Groups of 1 member (this should be an exceptional situation):

1. Only select one item from Step 4;
2. Students complete Steps 1, 2, 3, 4, 5 and 6. They do NOT do Step 7.

Submission requirements and format

READ the WQU Academic Policy on the Use of AI for a proper use of this tool. Instructors heavily penalize any improper use.

One team member submits on behalf of the entire group the following:

1. **1 PDF document***. This document will be around 15 pages and include all the relevant graphs and/or tables from your Notebook necessary to explain your conclusions. **Do not use the "See Notebook" as an answer**. The comments and relevant tables/graphs that support those comments MUST appear in the paper. Also make sure that the format of the document is uniform.

DO NOT FORGET to use the available Report Template and fill out the required information on the first page, then make sure to include the reports in the PDF along with the template.

2. **1 zipped folder including:**
   a. One Jupyter notebook with the code that contains clear section headings: **NOTE**: one and not multiple smaller notebooks! **MAKE SURE** that a pdf or html version of the .ipynb file is in there or **you will be penalized 5 points for failing to make a proper submission**.
      i. Your code MUST contain comments where you explain your procedures and choices.
      ii. One (1) pdf or html file version of the Jupyter notebook that contains the code, comments, **AND** results. The explanation of your results should be made in the paper.

** **Use Google Docs to collaborate** . Start by uploading the Report Template provided in the Course Overview. Once your report is completed, click File → Download → PDF Document (.pdf) to obtain the copy for your submission.*

*** **Use Google Colab or GitHub to collaborate** in completing the executable Python program.*

*The PDF files must be uploaded **separately** from the zipped folder that includes any other types of files. This allows Turnitin to generate a similarity report.*

Rubric

Your instructor will evaluate your group submission for GWP2 using the following rubric:

Quantitative Analysis (open-ended questions) 60 Points	Technical and Non-technical Reports 40 Points	Writing and Formatting 20 Points
The group is able to apply results, formulas, and their knowledge of theory to real-life finance scenarios by doing the following: ● Providing all the necessary information to support their arguments. ● Presenting arguments that reflect group discussion and research. ● Using authoritative references to support a position and provide updated information ● Concluding with practical takeaways for more insightful financial decision-making	Technical Reports contain 3 parts: 1) summary of key results; 2) interpretation of results; and 3) the recommended course of action that can reasonably follow from those results and interpretations. **Note**: Technical reports will include the technicalities of models, such as names, methods of estimation, parameter values, etc. and exclude generalities about the work done. It should NOT include names of Python code that were used.	A submission that looks professional should include: ● The axes, labels, and scales in graphs. ● No significant grammar errors or typos. ● Organized, well-structured, and easy-to-read document. ● Proper citations and bibliography using MLA format.
	Non-technical Reports contain 3 parts: 1) clear explanation of results; 2) the recommended course of action that follows; and 3) the identification of factors that impact each portfolio. **Note**: AVOID all references to model names, algorithms, and unnecessary details. Instead, focus on the investment decision.	

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